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Adolescent Health

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Objectives

- Identify the specific situations in which minors have the right to consent for their own health care in the state of California
- Recognize the HEADSS exam and identify appropriate questions for each section
- Identify Tanner Stages 1 – 5 for males and females
- Identify the vaccines that the CDC recommends that adolescents receive (usually at the age of 11-12 years)
- Recognize the goals of a Sports Physical
- Identify the factors that are protective against risk for the adolescent patient and those factors associated with increased risk

Adolescent Medicine

- Time of rapid biological and psychosocial change
- Generally few scheduled doctor visits
- Many seek health care only for acute illnesses or for required camp/sports participation

Purpose of Annual Preventive Health Visit

- Identify factors that place the adolescent at health risk
- Provide counseling regarding prevention of health risk behaviors
- Intervene when health risk factors are identified

Definitions

- ***Informed consent:*** describes the process during which the patient learns the risks and benefits of alternative approaches to management and authorizes a course of action proposed by the clinician
 - has both ethical and legal derivations
 - implies that the individual has the mental capacity to give informed consent
- ***Assent:*** agreement to undergo medical treatment obtained from an adult or child who is legally incompetent to consent
 - respects the decision-making skills of a minor by allowing them to participate in the decision

Consent

- Who is authorized to give consent for health care and whose consent is required?
- In general, U.S. law requires the consent of a parent before medical care can be provided to a minor - there are numerous exceptions:
 - Consent by someone other than a biologic parent
 - Foster parent, a juvenile court, a social worker, or probation officer
 - Emergency situations where care may be provided without prior consent to safeguard the life and health of the minor

Consent

- Specific legal provisions in particular states that allow minors to consent for specific areas of care
 - Some of these include:
 - **Contraceptive** care
 - **Pregnancy** related care
 - Dx and Tx for **sexually transmitted infections (STIs)**
 - Dx and Tx of either **HIV or AIDS**
 - Dx and Tx of reportable or **contagious diseases**
 - Examination and treatment related to **sexual assault**
 - Counseling and treatment for **drug or alcohol problems**
 - Counseling and treatment for **mental health issues**

Consent - Special Circumstances

- Many states have given consent rights to minors who have special status -These include:
 - **Emancipated** minors
 - **Married** minors
 - Minors in the **armed services**
 - Minors **living apart** from their parents
 - In some states "**mature minors**"





California Law

- Minors' right to consent to healthcare in the following situations:
 - **Contraceptive** services
 - **Prenatal care** (including abortion)
 - **STI/HIV** Services (minor must be at least 12)
 - Treatment for **alcohol and/or drug abuse** (minor must be at least 12 & doctor may notify parents)
 - Outpatient **mental health services** (minor must be at least 12 & doctor may notify parents)

Legal Limits of Confidentiality

- There are circumstances in which it is neither possible nor appropriate to maintain the confidentiality :
- The adolescent poses a **severe risk of harm to himself / herself or to others**
 - cases of **suspected physical or sexual abuse**
 - **specific laws** in some geographic areas that require parental notification in certain circumstances, even if the care is based on a teen's own consent
- When confidentiality must be breached for ethical or legal reasons, the adolescent should be informed

Interview

- Begin with both the patient and their parent in the room as you obtain the medical history
- Speak directly to the patient and to listen to their comments and concerns
 - Role model for parents
- For the final portion of the interview, ask the parent to step outside of the room so that you may speak with the patient in private
 - HEADSS Exam
- At the start of each visit it is important to inform the patient and their parent about confidentiality
- This does not extend to behaviors that place the adolescent or others at serious risk



Why Confidentiality is Important

- *The Needs of Clinical Practice*
 - Facilitate adolescents seeking necessary care and providing accurate, candid and complete health information
- *Developmental Needs*
 - Help support the adolescents' growing sense of privacy and **autonomy**
- *Safety Issues*
 - Protect the teen from humiliation and discrimination

Case

- 16-year-old female comes into the clinic with her mother for evaluation of acne and headaches. When you are taking the history, while her mom is out of the room, she discloses that she has a boyfriend, she is sexually active and he occasionally uses condoms. Her last menstrual period was two months ago.
- She is concerned about being pregnant. She would like to get a pregnancy test. She also would like to get contraceptive pills if she is not pregnant. She asks that you do not tell her mother that she is sexually active or that she is getting a pregnancy test.

Case Questions

- Can you do a pregnancy test without the mother's permission?
- Can you prescribe oral contraceptives without the mother's permission?



Case Discussion

- Many countries and local states or provinces allow for confidential care of pregnancy in teens
 - This often includes both consent and the confidentiality of that information

Case Discussion Continued

- It's important to explore several issues first
 - What are her reasons for not wanting to disclose and share this information with her parents?
 - In some cases, after a discussion of the issues, teens are willing to share information with their parents.
 - What is the nature of her relationship with her boyfriend and is it consensual
 - Does she have any symptoms of a sexually transmitted infection?
 - If she is pregnant, has she thought about what she might do and who she might share that information with?

History – Special Considerations

- Past Medical History
 - Recurrent illness
 - Chronic diseases
 - Current medications
 - **OTC, herbal, inhaled, oral contraceptives, injectables**
 - Allergies
 - History of hospitalizations and surgeries
 - History of injuries and trauma
 - **Alcohol and personal violence**
 - Immunization history

History – Special Considerations

- Family History
 - Cardiovascular disease
 - MI or stroke before 55, sudden death, arrhythmias, HTN, high cholesterol and triglycerides
 - Asthma
 - Clotting disorders
 - Diabetes
 - Alcohol and drug abuse
 - Psychiatric disorders
 - Including suicide ideation/attempt

History – Special Considerations

- Review of Systems: Nutrition
 - Calcium, iron, fat intake
 - Vegetarian diet
 - Weight loss methods
 - History of weight loss or gain
 - Body image



History – Special Considerations

- Review of Systems: Gynecologic
 - Age of menarche
 - Frequency and regularity of menses
 - Duration of menses
 - Dysmenorrhea
 - Including medications taken
 - Last menstrual period
 - Tampon or pad use



What do I ask?

- HEADDS Exam for teens
 - Ask parents and siblings to leave the room
 - Discuss confidentiality (everything will be confidential with the exception of abuse and potential harm to self or others)



HEADSS Exam

- Home
 - Who do you live with? Do you get along with everyone at home? Any major fights or disagreements? Do you feel safe at home?
- Education
 - What grade are you in? What do you want to do when you finish with school? Do you feel safe at school? Do you have friends at school? Do you need extra help with your classes? What are your grades like?
- Activities
 - What do you do when you are not in school? Talk about internet sites and safety. Sports safety. Do you have a job?

HEADSS Exam

- Drugs/Alcohol/Cigarettes

- Do you or any of your friends drink alcohol? How much? How often? Do any drugs? List specific examples. Smoke cigarettes? How much? How often? Do you have someone safe that you can call if you get into a situation with drugs or alcohol and you need a ride home?

- Depression

- What do you do when you are depressed? Do you have someone that you can talk to?

- Suicide

- Have you ever tried to commit suicide? Have you ever purposefully hurt yourself. Do you feel like hurting yourself or killing yourself now? How about someone else?

- Sex

- Do you have a boyfriend, girlfriend or both? Have you ever? Have you ever had sex? Get specific about the different kinds of sexual activities. Do you use protection? What kind? How often? How many partners have you had? Have you ever had an STI? Ever been tested – when, what were the results?

Physical Exam – Special Considerations

- Often more uncomfortable with exam than adults/children
- Start with least embarrassing areas, saving most sensitive areas for last
- Talk through the examination so patient knows what to expect
 - Presence of chaperone is recommended for sensitive exams.

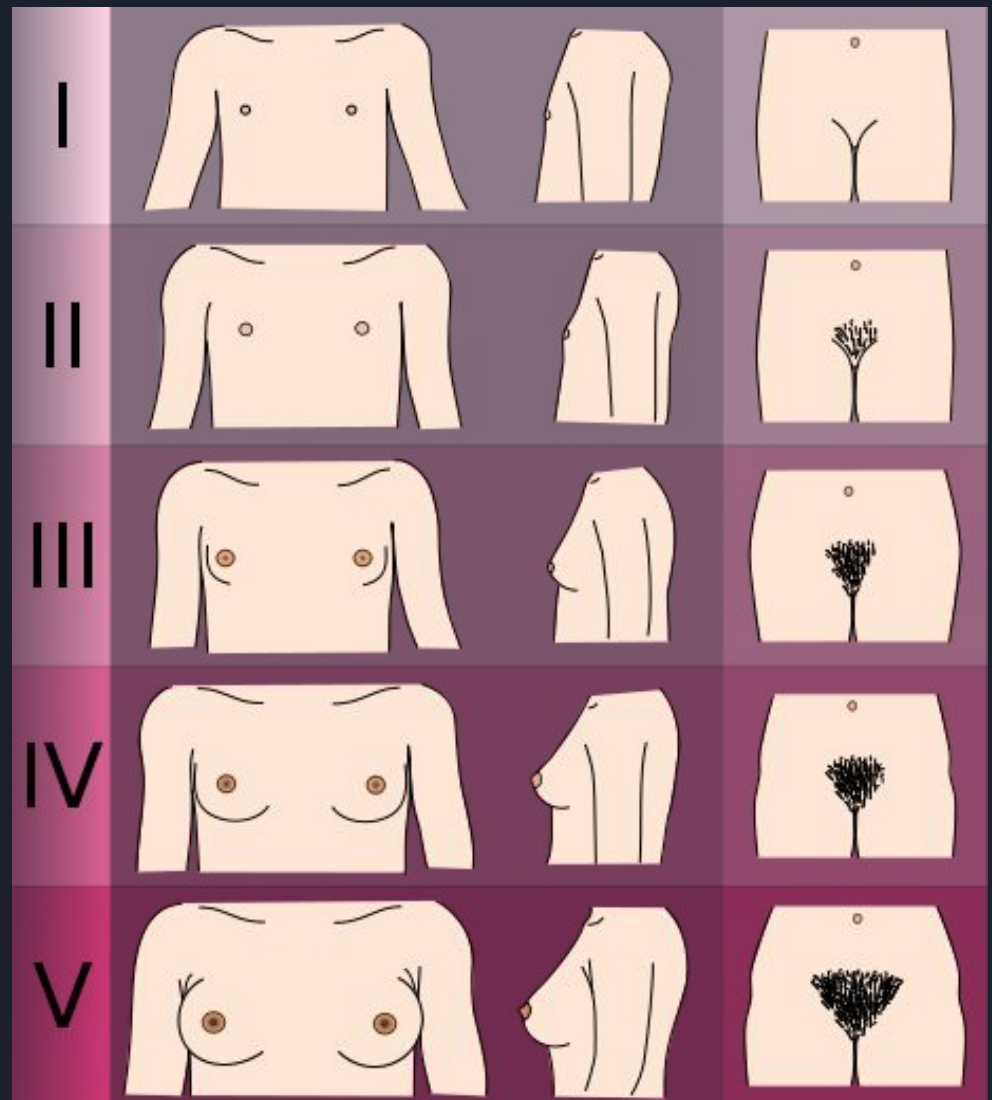
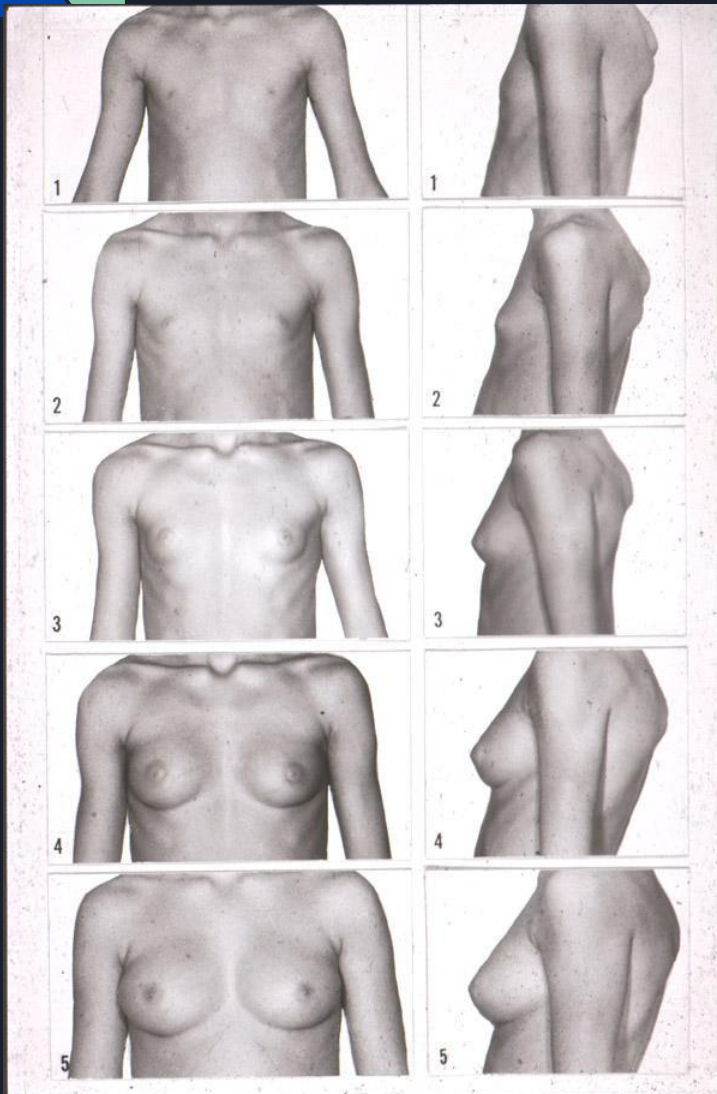
Physical Exam – Special Considerations

- **BMI** (for children and adolescents <18, obesity is >95% for gender and age)
 - If patient has lost 10% or more of body weight, or if they are <5% for age and gender, consider eating disorder or underlying organic disease
- **Blood pressure** (for children and adolescents <18, elevated BP is >95% for gender and age)
- Assess breasts and genitalia for **tanner stage** (and gynecomastia)
- **Hernias**
- **Pap smear** at 21 years of age (no longer within 3 years of sexual activity)
- **Scoliosis Screen** with Adam's forward bend test

Sexual Maturation

- Measured by Sex Maturity Ratings, or Tanner Stages
- Range from Tanner Stage 1 (prepubertal) to Tanner Stage 5 (fully developed)
- Early Sexual Development is Tanner Stage 2
- **Precocious Puberty**- development of secondary sexual characteristics before 8 y.o. in females and before 9 y.o. in males
 - African American females tend to develop ~ 1 year earlier than caucasian females
- **Adrenarche** – Pubic Hair (adrenal hormones)
- **Thelarache** – Breast Development (estrogen)

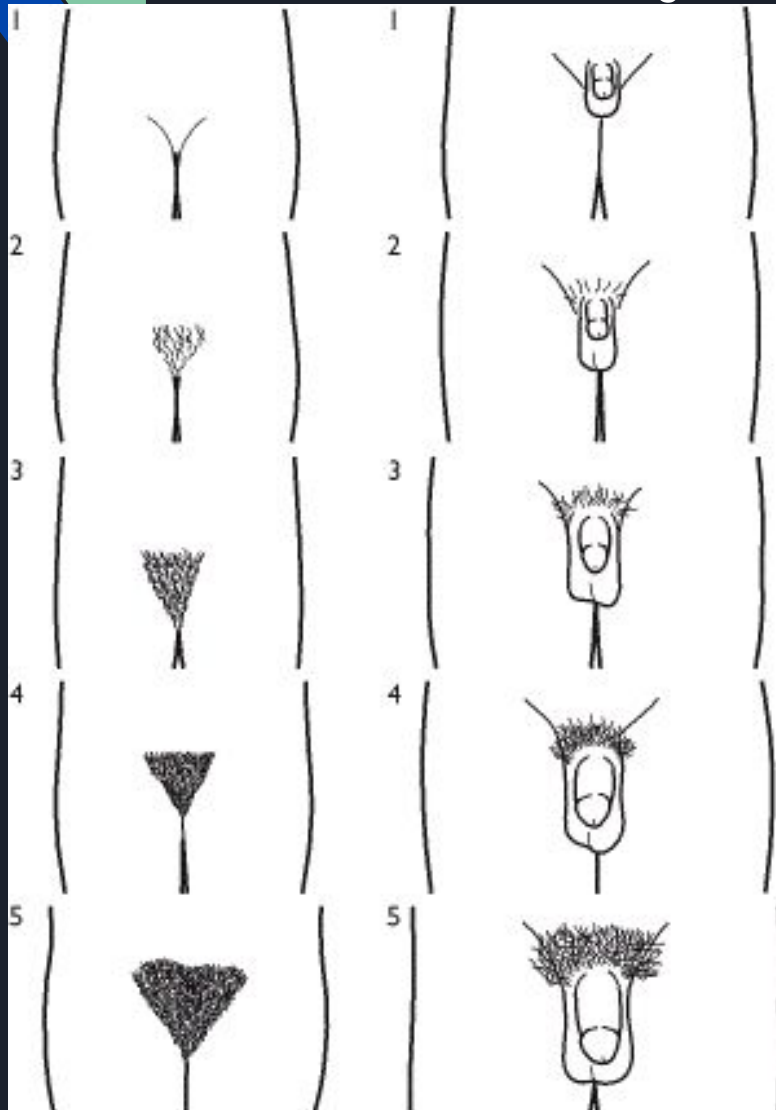
Sexual Maturity Rating Tanner Stage



Tanner Stage – Breasts (Female)

- **Tanner I** - no glandular tissue; areola follows the skin contours of the chest (prepubertal) [typically age 10 and younger]
- **Tanner II** - breast bud forms, with small area of surrounding glandular tissue; areola begins to widen [10-11.5 yr]
- **Tanner III** - breast begins to become more elevated, and extends beyond the borders of the areola, which continues to widen but remains in contour with surrounding breast [11.5-13 yr]
- **Tanner IV** - increased breast size and elevation; areola and papilla form a secondary mound projecting from the contour of the surrounding breast [13-15 yr]
- **Tanner V** - breast reaches final adult size; areola returns to contour of the surrounding breast, with a projecting central papilla. [15+ yr]

Sexual Maturity Rating



I		3 ≤2,5
II		4 2,5-3,2
III		10 3,6
IV		16 4,1-4,5
V		25 >4,5



Tanner Stage – Pubic Hair (Male and Female)

- **Tanner I** - no pubic hair at all [typically age 10 and younger]
- **Tanner II** - small amount of long, downy hair with slight pigmentation at the base of the penis and scrotum or labia majora [10–11.5 yr]
- **Tanner III** - hair becomes more coarse and curly, and begins to extend laterally [11.5–13 yr]
- **Tanner IV** - adult-like hair quality, extending across pubis but sparing medial thighs [13–15 yr]
- **Tanner V** - hair extends to medial surface of the thighs [15+ yr]

Tanner Stage – Genitalia (Males)

- **Tanner I** - prepubertal (testicular volume less than 1.5 ml; small penis of 3 cm or less) [typically age 9 and younger]
- **Tanner II** - testicular volume between 1.6 and 6 ml; skin on scrotum thins, reddens and enlarges; penis length unchanged [9-11 yr]
- **Tanner III** - testicular volume between 6 and 12 ml; scrotum enlarges further; penis begins to lengthen to about 6 cm [11-12.5 yr]
- **Tanner IV** - testicular volume between 12 and 20 ml; scrotum enlarges further and darkens; penis increases in length to 10 cm and circumference [12.5-14 yr]
- **Tanner V** - testicular volume greater than 20 ml; adult scrotum and penis of 15 cm in length [14+ yr]

Components of Health Maintenance Counseling

- Normal Growth and Development
- Peer Pressure
- Nutrition
- Exercise
- Injury Prevention
- Violence
- Sexuality
- Tobacco Use
- Alcohol and Other Drugs
- Emotional and Psychological Health
- Parents



Immunizations

- CDC currently recommends these 3 vaccines at 11-12 years of age (or soon afterwards):
 - **Tdap** (tetanus - diphtheria booster and pertussis booster)
 - **Meningococcal** vaccine
 - Booster at 16 years
 - **Human papillomavirus** vaccine (2 or 3 shot series)
 - Approved for females age 9 – 26 years
 - In January 2010 was approved for males aged 9-18 years
 - Now option to give 2 doses 6 months apart if starting before age 15

Immunizations



- If they did not receive all recommended doses when they were younger, teens should also receive:
 - **Hepatitis B** series
 - **Polio** series
 - **Measles-mumps-rubella (MMR)** series
 - **Varicella (chickenpox)** series
 - A second catch-up varicella shot is now recommended for children, adolescents, and adults who have previously received one dose
- Flu vaccine annually and COVID vaccine also recommended.

Labs/Imaging

Consider as needed (not exhaustive and may not be indicated for all patients):

- 1) CBC (especially if menstruating), (BUN/Cr/Electrolytes)
- 2) Obesity panel: A1C, TSH, fasting glucose, AST, ALT, lipid panel
- 3) STI screening: GC/CT, HIV (screen once before 18), RPR, ?Hepatitis panel
- 4) TB screen: Quantiferon Gold blood test
- 5) Beta HcG (urine pregnancy test)
- 6) Imaging as indicated for concerns

Sports Physicals

Pre-participation Evaluation

- **Goals:**
 - **Prevent sudden death**
 - **Help athlete participate safely**
 - **Identify injuries and conditions that may affect participation or affect the overall well-being of the teen**

Sports Physicals Pre-participation Evaluation



- Important history questions:
 - Personal or family history of Cardiovascular disease
 - MI or stroke before 55, sudden death, arrhythmias, HTN, high cholesterol and triglycerides
 - History of **concussions**
 - Biggest risk factor = hx of previous concussion
 - Additive effect of head trauma
 - **Shortness of breath and/or cough after exercise**
 - Desired **weight/body image**
 - Number of **menstrual periods** in last 12 months
 - Amenorrhea or oligomenorrhea put athletes at risk for stress fractures and loss of bone mineral density
 - **Orthopedic history** (musculoskeletal injury, rehab)

Sports Physicals Preparticipation Evaluation

- Blood pressure
- Pulse
- Physical exam with thorough **musculoskeletal exam**
- Counseling regarding **overuse injuries**





Resources for Sports Physicals (AAP)

History:

https://downloads.aap.org/AAP/PDF/PPE_History-form-7-30-21.pdf

Physical:

https://downloads.aap.org/AAP/PDF/PPE_Physical-Exam-7-30-21.pdf

Apophysites

- Among the most common overuse injuries to occur in adolescents
- **Apophyses** = traction growth plates that usually correspond to bony prominences at tendinous attachments
- Usually due to **repetitive microtrauma** of fibrocartilage

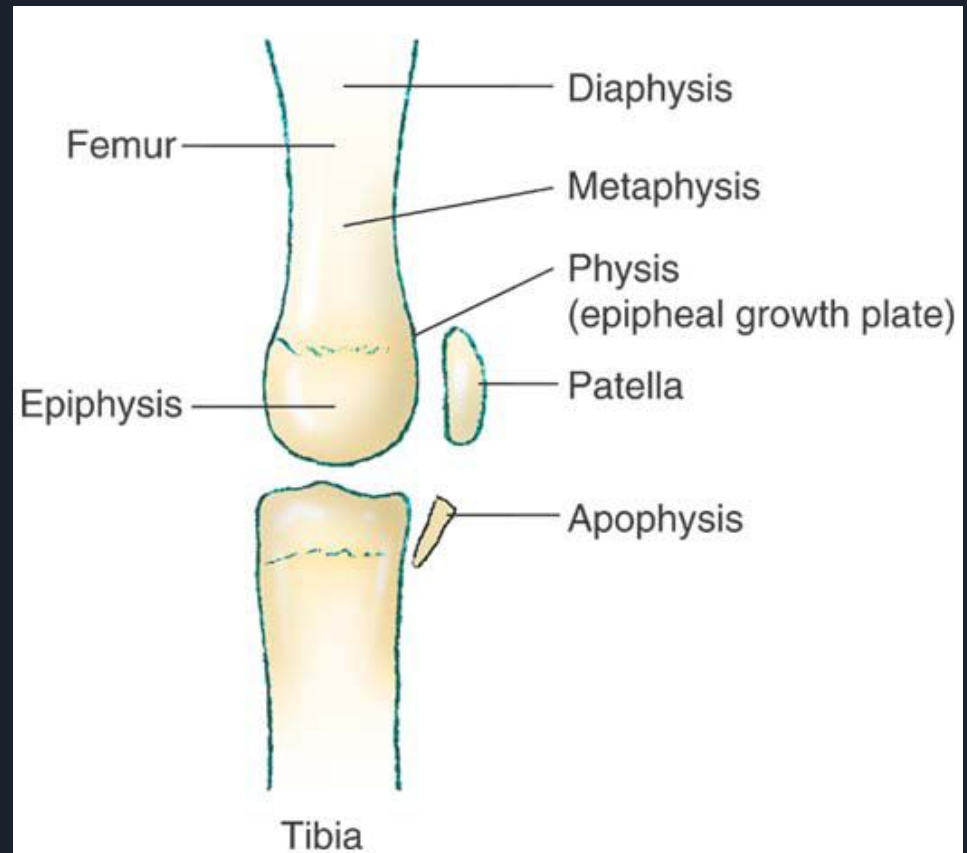


Figure 219-1 The apophysis is a traction growth plate and is located at the tibial tubercle for the patellar tendon. The pressure physis is located between the epiphysis and metaphysis on long bones.

Mosby items and derived items © 2005 by Mosby, Inc.

Most Common Apophysites

- **Calcaneal Apophysitis**

- Calcaneal apophysis closes ~ 17 years
- Soccer players
- Achilles tendonitis

- **Osgood-Schlatter Disease**

- Usually at 11- 15 yrs, females more common
- Swelling and pain at tibial tubercle
- Excessive running, jumping, hard surfaces, tight hamstrings, midfoot hyperpronation



- **Iliac Apophysitis**

- Running athletes with dull, achy hip pain with activity
- Tenderness to iliac crests (usually ASIS)

- **Treatment** = rest, +/- NSAIDs, ice, stretching and strength exercises once pain has resolved



Figure 219-2 Hyperpronation. This young athlete has marked bilateral ankle valgus and midfoot pronation. She is at risk for a variety of overuse syndromes with this degree of hyperpronation.

Adolescent Risk

- Risk taking in adolescence is common
- Lack life experience
- May feel invincible
 - Adolescent mental health
- Have pressure from peers, friends, family and society
- May not have the ability to think abstractly or far into the future (frontal lobe not fully developed)
- When counseling teens you may have more luck focusing on the immediate & superficial consequences
 - Smoking will stain your teeth and make you smell badly
 - As opposed to “smoking causes cancer”



Evaluating Risk

- Do you and/or your friends participate in activities that you feel are risky? What are they?
- Do you smoke (cigarettes/marijuana)?
- How do you drink alcohol?
- Have you tried drugs or prescription medications that weren't prescribed to you?
- Are you sexually active? How many partners have you had?
- When you drive cars, boats, ATVs, etc, are you ever under the influence of drugs or alcohol?
- Have you ever gotten into the car with a driver who was under the influence of drugs or alcohol?
- Do you use a seatbelt in the car? How often?
- Do you wear a bike helmet? How often?
- Are there guns in your home? Are they locked up?
- Do you feel safe with the person you are seeing?



Fast Facts for US:

<https://www.cdc.gov/nchs/fastats/adolescent-health.htm>

Health status

- Percent of adolescents aged 12–17 years who are in fair or poor health: 3.8% (2022)
- Percent of adolescents aged 12–17 years who missed 11 or more days of school in the past 12 months because of illness, injury, or disability: 9.9% (2022)

Obesity:

- Percent of adolescents aged 12-19 years with obesity: 22.2% (2017-March 2020)

Smoking and alcohol use

- Percent of adolescents aged 12–17 years who smoked cigarettes in the past month: 2.3% (2019)
- Percent of adolescents aged 12–17 years who used alcohol in the past month: 9.4% (2019)



Fast Facts for US:

<https://www.cdc.gov/nchs/fastats/adolescent-health.htm>

Access to health care

- Percent of adolescents aged 12–17 years with a usual source of health care: 96.4% (2022)
- Percent of adolescents aged 12–17 years without health insurance: 4.6% (2022)

Mortality

- Number of deaths for adolescents aged 15–19 years: 13,407
- Deaths per 100,000 population for adolescents aged 15–19 years: 62.2

Leading causes of deaths among adolescents aged 15–19 years:

- Accidents (unintentional injuries)
- Homicide
- Suicide

Protective Factors for Teens

- Strong verbal and communication skills
- Easy temperament
- Ability to solve problems
- Sense of humor
- Empathy
- Perspective-taking skills
- Spirituality
- Strong family functioning and communication
- Extra-family support from community and/or other adults



Factors Associated with Increased Risk

- Lack of close relationships with parents
- Minimal parental supervision
- Family chaos
- School transitions
- Peers who participate in risky behaviors
- Illicit substance use availability
- Male gender
- Lack of concern about consequences of behavior

Risk Reduction Messages

- Don't drink alcohol or use drugs
- Don't smoke cigarettes
- Practice sexual abstinence
- Practice safer sex if you are sexually active
- Use a seat belt in the car
- Use a bike helmet or motorcycle helmet
- Be aware that having a gun in the home always poses risks
- Avoid people who have guns illegally
- Learn how to solve conflicts without violence
- Use protective equipment for sports
- Learn to swim
- Wear life jackets for boating and water sports
- Learn first aid and CPR

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